

University of Wisconsin
SCHOOL OF MEDICINE
AND PUBLIC HEALTH

BM / STAT- 768
St at i st i cal Met hods f or Medi cal I mage Anal ysi s and Hi gh Di mensi onal Dat a

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University of Wisconsin-Madison

https://github.com/laplcebeltrami/BMI768

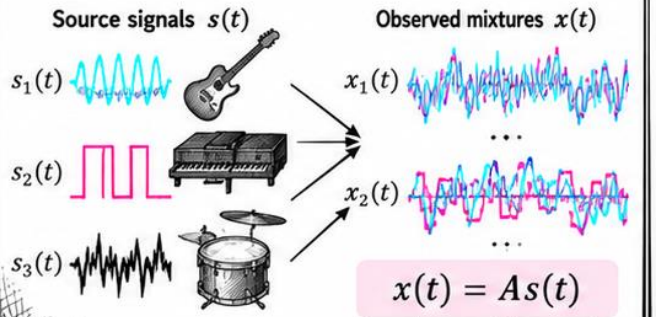
Spectral Learning

Learning outcome

1. Learn spectral embedding
2. Learn how to simulate spectral expansion
3. Learn inference through spectral expansion

Independent Component Analysis

Separate mixed signals into statistically independent sources.



1. The ICA Model

Observed data are linear mixtures of independent sources.

$$\underset{(m\times 1)}{x(t)} = \underset{(m\times n)}{A} \cdot \underset{(n\times 1)}{s(t)}$$

$x(t)$: observed mixtures
$s(t)$: source signals (independent)
A : unknown mixing matrix

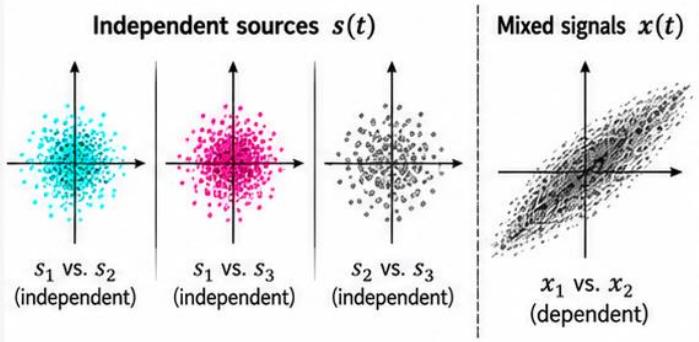
Goal: find an unmixing matrix W such that

$$\hat{s}(t) = Wx(t) \approx s(t)$$

with statistically independent components $\hat{s}_i(t)$.

2. Independence

Source signals are statistically independent (non-Gaussian).



ICA relies on the non-Gaussianity and independence of the sources.

3. Estimation (e.g., FastICA)

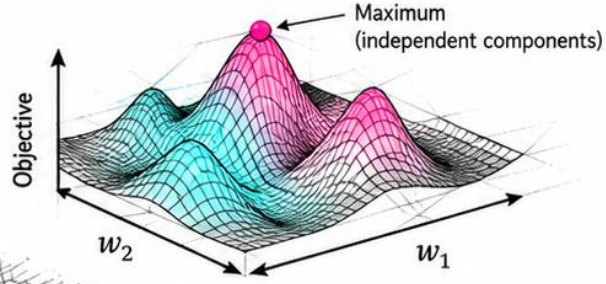
Find W by maximizing non-Gaussianity (e.g., kurtosis or negentropy).

$$\max_W \sum_{i=1}^{n} G(\hat{s}_i(t))$$

subject to $WA = I$
(up to scale and permutation)

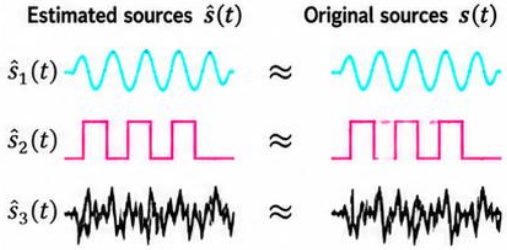
Examples of contrast functions:

$G(u) = \frac{u^4}{4}$ (kurtosis)
$G(u) = -\log\cosh(u)$
$G(u) = -H(\hat{s}_i)$ (negentropy)



4. Recovered Sources

Estimated components $\hat{s}(t)$ recover the original independent signals (up to scaling and permutation).

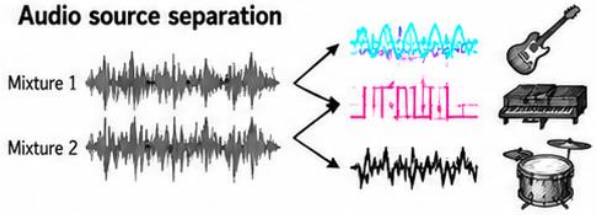


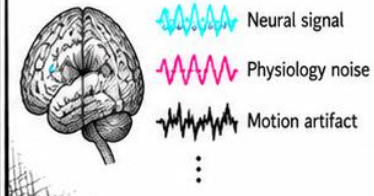
ICA recovers the sources up to permutation and scaling:

$$\hat{s}_i(t) \approx \alpha_i s_{\pi(i)}(t)$$

5. Applications

ICA is widely used to separate mixed signals in many fields.





Other applications

EEG, MEG, speech processing, image analysis, financial data, and more ...

Find the hidden independent sources behind the mixtures.

Dictionary Learning

Learn a set of basis elements (dictionary) that sparsely represent data.



$$x_i \approx D\alpha_i$$

D : dictionary
α_i : sparse code (few nonzero)

Data samples

$X = [x_1, x_2, \ldots, x_N]$

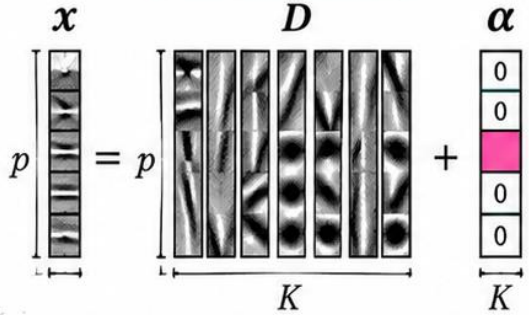
$x_i \in \mathbb{R}^p$

Learn D and α_i from the data itself.

1. Sparse Representation

Each signal is approximated by a sparse linear combination of dictionary atoms.

$$x \approx D\alpha$$



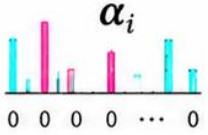
$$\|\alpha\|_0 \le s \quad (s \ll K)$$

2. Learn the Dictionary

Find D and sparse codes $\{\alpha_i\}$ that minimize reconstruction error with sparsity constraint.

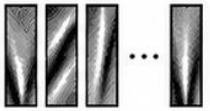
$$\min_{D,\{\alpha_i\}} \sum_{i=1}^{N} \|x_i - D\alpha_i\|_2^2 + \lambda \|\alpha_i\|_1$$

Update sparse codes (fix D)



Sparse coding (e.g., Lasso)

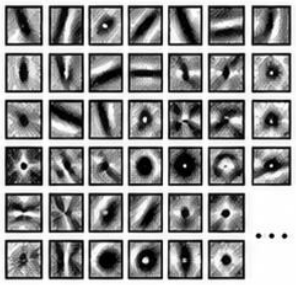
Update dictionary (fix $\{\alpha_i\}$)



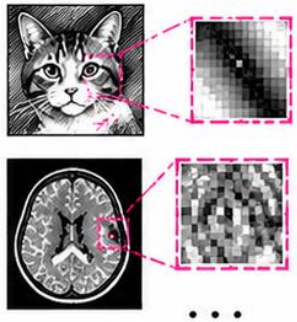
Dictionary update (e.g., K-SVD)

3. Learned Dictionary Atoms

Atoms are learned from the data and adapt to its structure (e.g., edges, textures, patterns).

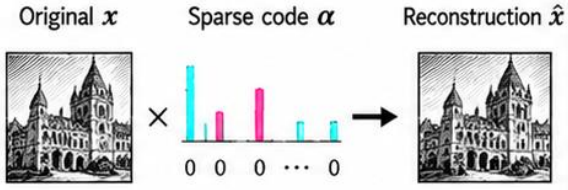


Learned dictionary D



4. Reconstruction with Few Atoms

A signal can be well reconstructed using only a few dictionary atoms.

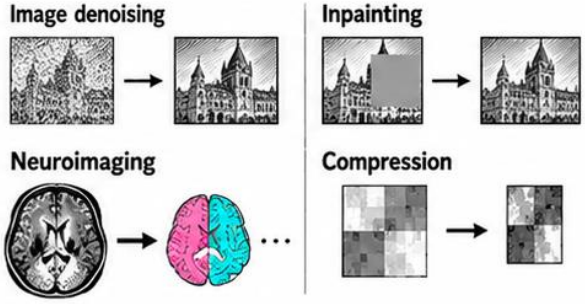


$$\hat{x} = D\alpha$$

Accurate reconstruction from a sparse code.

5. Applications

Dictionary learning provides compact, interpretable representations for many data types.

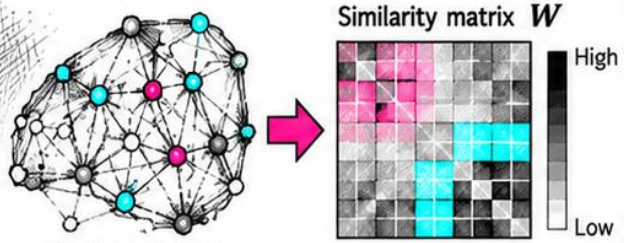


Learn the right basis from the data.
Sparser. Cleaner. More interpretable.

Spectral embedding

Spectral Learning

Learn data structure from eigen-decomposition of a matrix (L or A).



Data as a graph
(or samples in space)

$$W_{ij} = \text{similarity}(x_i, x_j)$$

(e.g., kNN, Gaussian kernel)

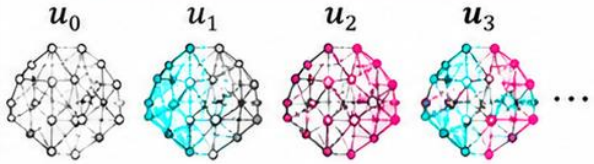
1. Spectral Decomposition

Compute the eigen-decomposition of the graph Laplacian or affinity matrix.

$$L = D - W$$

$$Lu_k = \lambda_k u_k$$

$$0 = \lambda_0 \le \lambda_1 \le \cdots \le \lambda_{n-1}$$

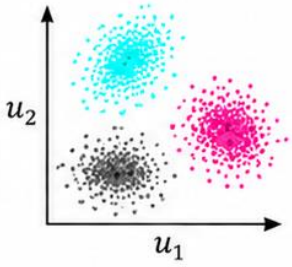


Low frequency (global) — High frequency (local)

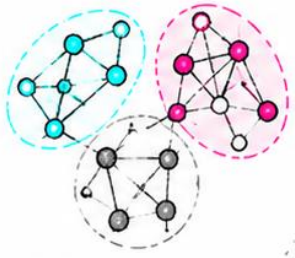
2. Learning from the Spectrum

Use leading eigenvectors (spectral embedding) for downstream learning tasks.

Dimensionality reduction



Clustering / classification



Manifold learning (spectral embedding)

$$Y = [u_1, \ldots, u_k] \in \mathbb{R}^{n \times k}$$

Learn geometry from the spectrum.

3. Variants and Models

Spectral learning appears in many models.

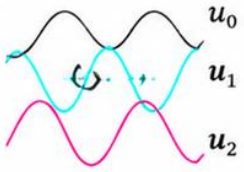
Spectral clustering

$$\min_F \ \text{Tr}(F^T L F)$$
$$F^T F = I$$



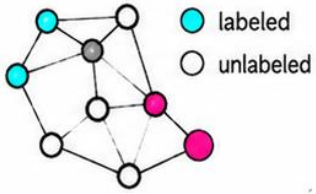
Graph signal processing

$$x = \sum_k c_k u_k$$



Semi-supervised learning

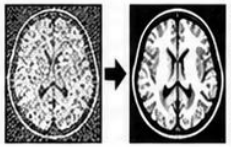
$$\min_f f^T L f + \mu \sum_{i \in L} (f_i - y_i)^2$$



4. Applications

Widely used in science and engineering.

Image analysis



Denoising
Segmentation

Brain networks



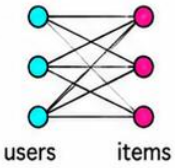
Functional connectivity
Network parcellation

Single-cell / genomics



Trajectory inference
Cell type discovery

Recommendation



Link prediction
Matrix completion

Spectral learning: simple math, powerful representations.

Situating the default-mode network along a principal gradient of macroscale cortical organization

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Edited by Peter L. Strick, University of Pittsburgh, Pittsburgh, PA, and approved September 9, 2016 (received for review May 27, 2016)

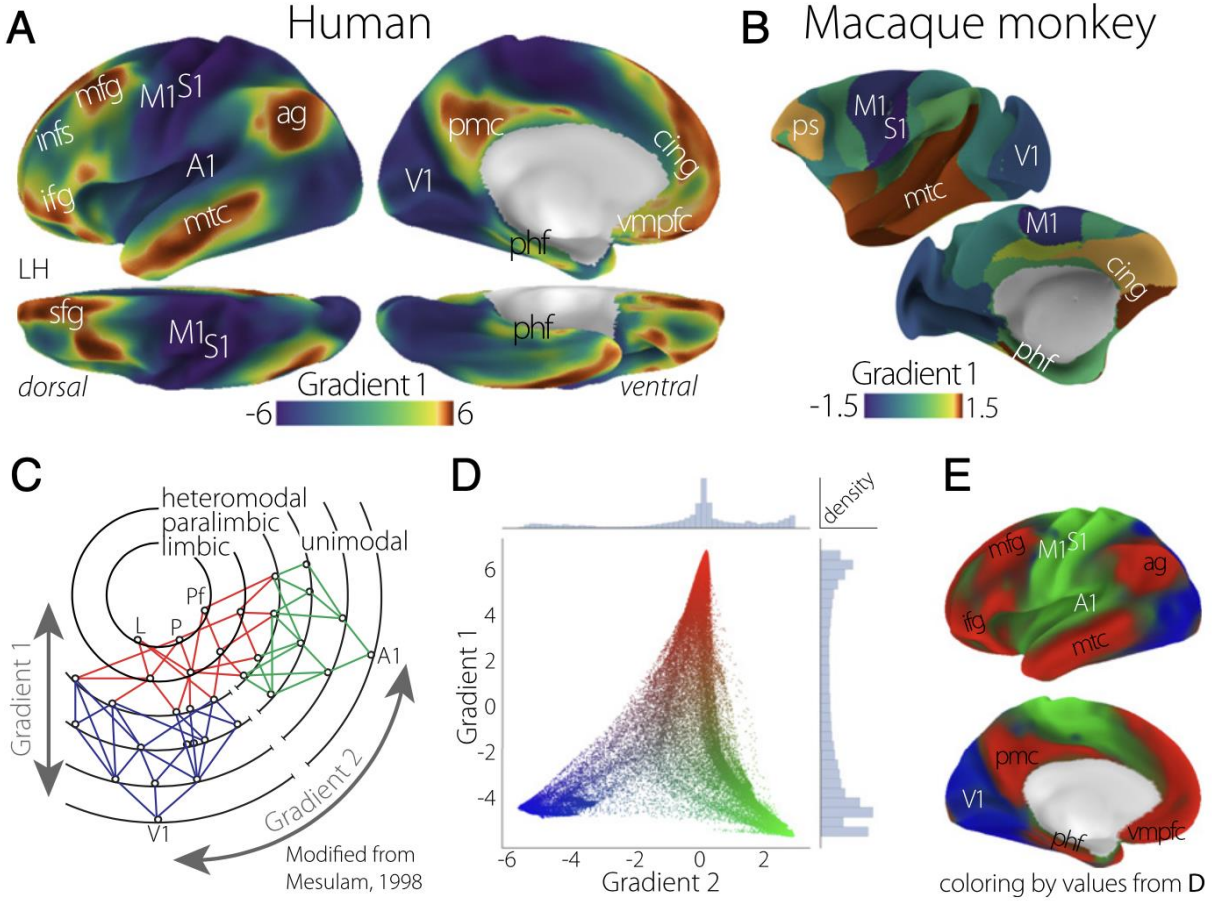


Fig. 1. The principal gradient of connectivity in both the (*A*) human and (*B*) macaque monkey cortices shows a spectrum between unimodal regions (dark blue) and transmodal regions (sienna), which in the human cortex, peaks in regions corresponding to the DMN. The proximity of colors can be interpreted as greater similarity of connectivity patterns. (*C*) The illustration of connectivity organization suggested by Mesulam (23) proposes a hierarchy of processing from distinct unimodal areas to integrative transmodal areas. Labels Gradient 1 and Gradient 2, which were not included in the original figure, correspond to the results in *D*. Modified from ref. 23. (*D*) A scatter plot of the first two connectivity embedding gradients. Gradient 1 extends between primary sensorimotor and transmodal regions (red). Gradient 2 separates somatomotor and auditory cortex (green) from visual cortex (blue). Histograms depicting the distribution of values are presented on the respective axes. (*E*) Colors from the scatter plot are presented on the cortical surface for anatomical orientation. A1, primary auditory; ag, angular gyrus; cing, anterior cingulate cortex; ifg, inferior frontal gyrus; infs, intermediate frontal sulcus; L, limbic; M1, primary motor; mfg, middle frontal gyrus; mtc, middle temporal cortex; P, parietal; Pf, prefrontal; phf, parahippocampal formation; pmc, posteromedial cortex; ps, principal sulcus; S1, primary somatosensory; sfg, superior frontal gyrus; V1, primary visual; vmpfc, ventromedial prefrontal cortex.

Project: Hodge spectral network gradient

Check for updates

Gradients of structure–function tethering across neocortex

Bertha Vázquez-Rodríguez[a], Laura E. Suárez[a], Ross D. Markello[a], Golia Shafiei[a], Casey Paquola[a], Patric Hagmann[b], Martijn P. van den Heuvel[c], Boris C. Bernhardt[a], R. Nathan Spreng[a], and Bratislav Misic[a,1]

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Edited by Marcus E. Raichle, Washington University in St. Louis, St. Louis, MO, and approved September 10, 2019 (received for review February 26, 2019)

The white matter architecture of the brain imparts a distinct signature on neuronal coactivation patterns. Interregional projections promote synchrony among distant neuronal populations, giving rise to richly patterned functional networks. A variety of statistical, communication, and biophysical models have been proposed to study the relationship between brain structure and function, but the link is not yet known. In the present report we seek to relate the structural and functional connection profiles of individual brain areas. We apply a simple multilinear model that incorporates information about spatial proximity, routing, and diffusion between brain regions to predict their functional connectivity. We find that structure–function relationships vary markedly across the neocortex. Structure and function correspond closely in unimodal, primary sensory, and motor regions, but diverge in transmodal cortex, particularly the default mode and salience networks. The divergence between structure and function systematically follows functional and cytoarchitectonic hierarchies. Altogether, the present results demonstrate that structural and functional networks do not align uniformly across the brain, but gradually uncouple in higher-order polysensory areas.

connectome | structure–function | cortical gradient

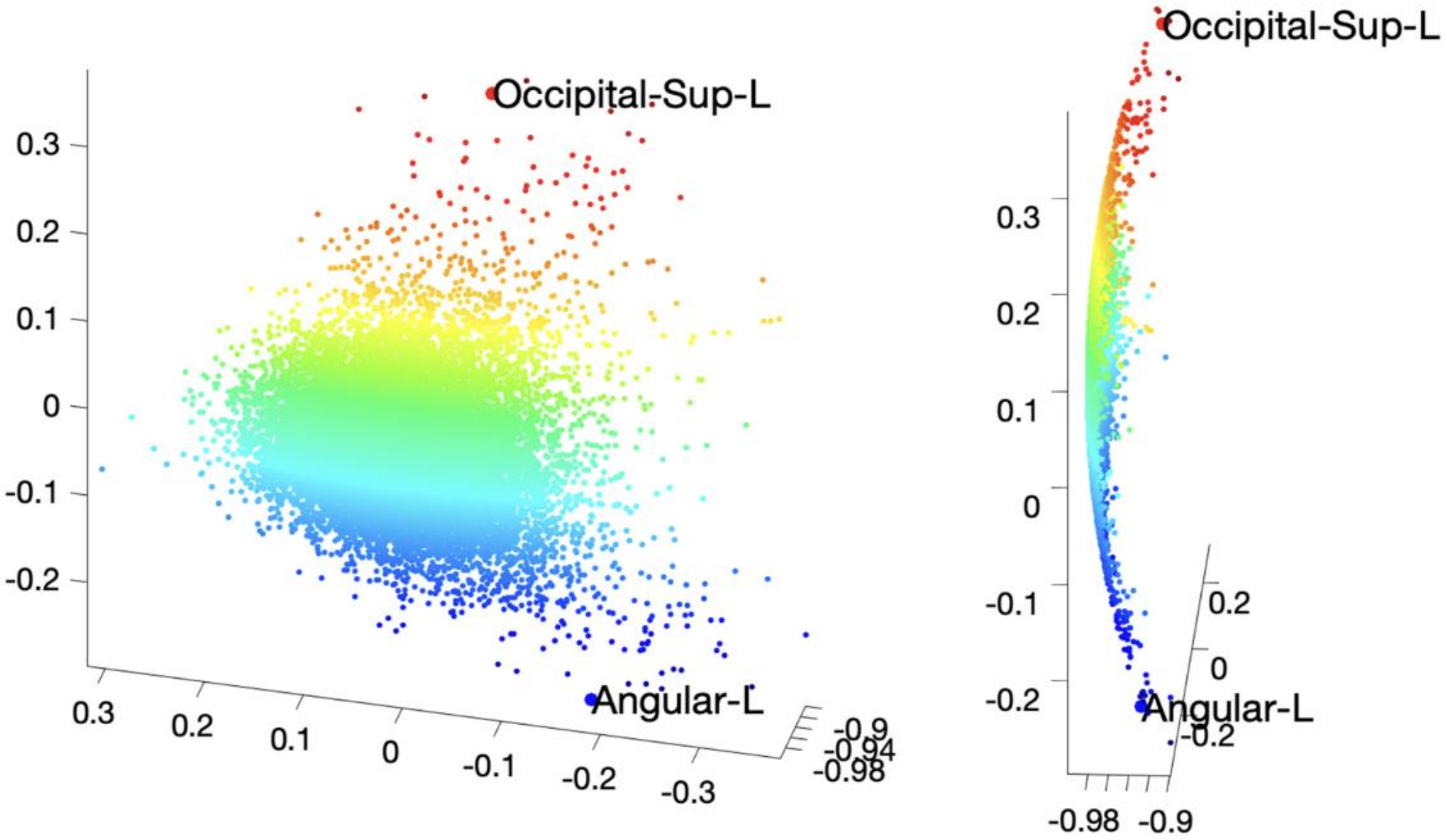
ciations with intracortical myelination (21) and laminar differentiation (22). A prominent account posits that rapid evolutionary expansion of association cortices effectively "untethers" polysensory regions from molecular signaling gradients and canonical sensory-motor activity cascades, resulting in fundamentally different structure–function relationships along the unimodal–transmodal hierarchy (23). Altogether, this work opens the possibility that structure and function may not be related in exactly the same way across the whole brain, but potentially converge or diverge in specific areas.

Here we address the relationship between structure and function by focusing on connection profiles of individual brain regions. We first reconstruct structural and functional networks from diffusion MRI (dMRI) and resting-state functional MRI (fMRI) in a cohort of 40 healthy participants. We then apply a simple multilinear model that uses information about a region's geometric and structural network embedding to predict its functional network embedding. The method allows us to ask how closely structure and function correspond in individual regions and the extent to which this correspondence reflects affiliation with cognitive systems, cytoarchitecture, and functional hierarchies.

NEUROSCIENCE

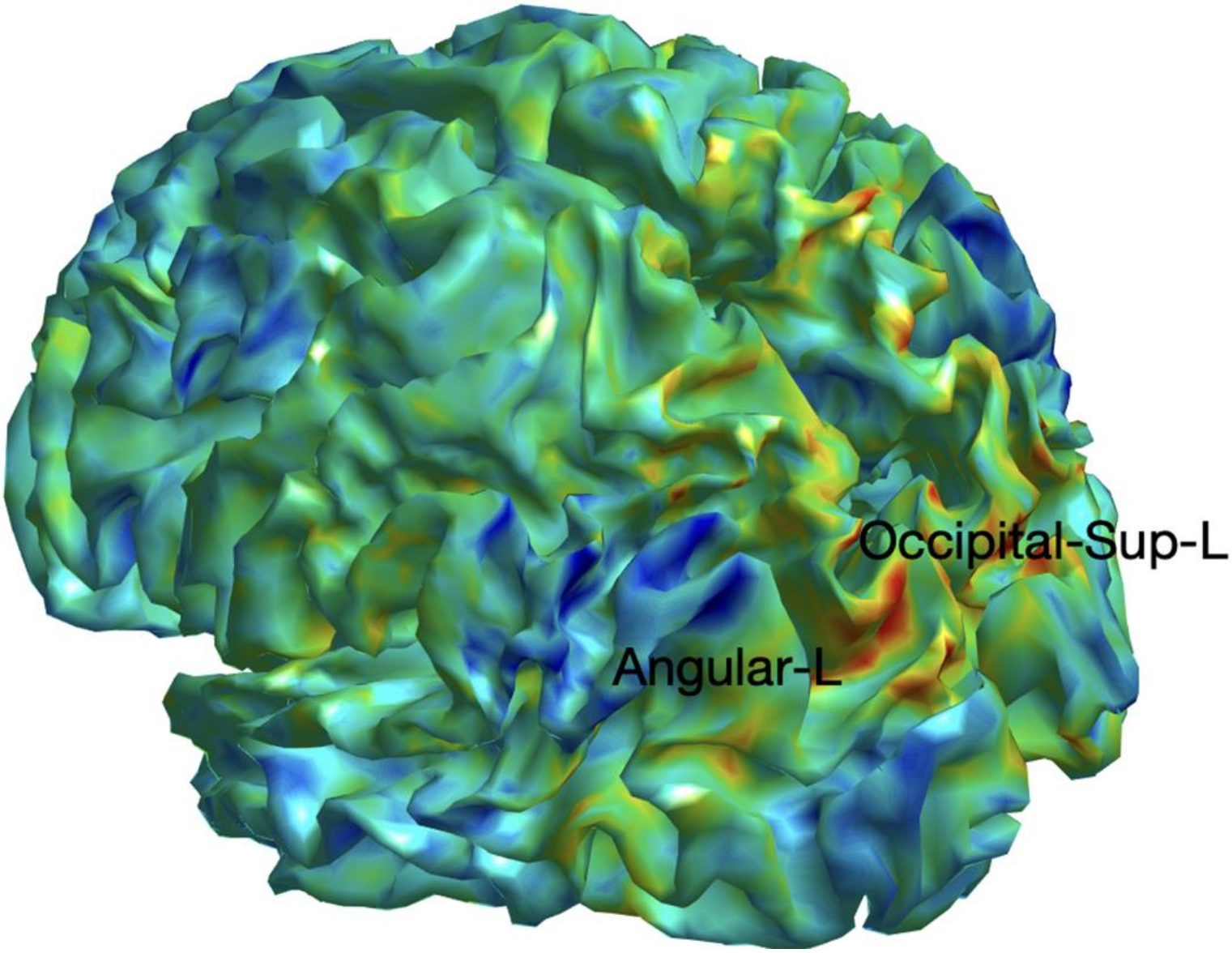
Exploratory descriptive method → How to quantify?

Largest network gradient from spherical-MDS



Project: Spherical-MDS of brain connectivity

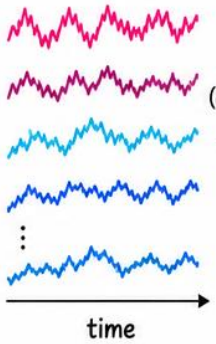
Largest network gradient from spherical-MDS



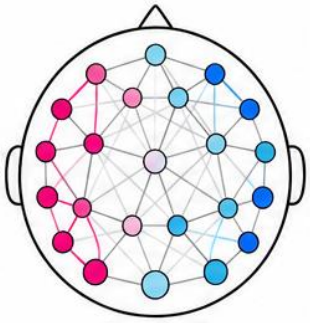
Spectral embedding using 1-

1. From brain signals to a graph

EEG signals
from many electrodes



Functional
connectivity
(e.g. correlation)



Graph G
(nodes: electrodes,
edges: functional connections)

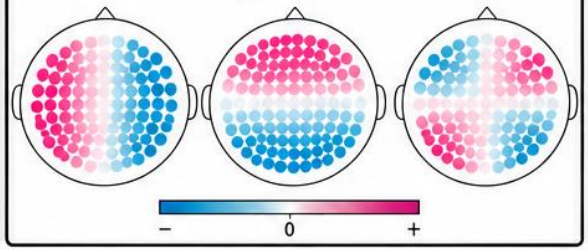
2. Spectral embedding with 0-Laplacian (scalar)

Graph Laplacian $L = D - W$ (D: degree, W: connectivity)

Eigen-decomposition

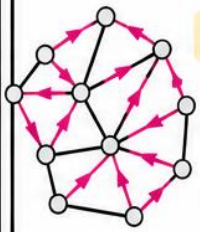
$$L\phi_k = \lambda_k \phi_k$$ (small λ_k: smooth global patterns)

ϕ_1 (Fiedler) ϕ_2 (Gradient) ϕ_3



3. From edges to 1-Hodge Laplacian

Define a flow on edges (directed interactions)



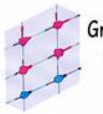
Edge flow X
(on each connection)

1-Hodge Laplacian

$$\Delta_1 = B_1^T B_1 + B_2 B_2^T$$

B_1 : node-edge incidence (gradient)

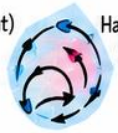
B_2 : edge-triangle incidence (curl)



Gradient flow
$(B_1^T B_1)$



Curl flow
$(B_2 B_2^T)$



Harmonic flow
(null space)

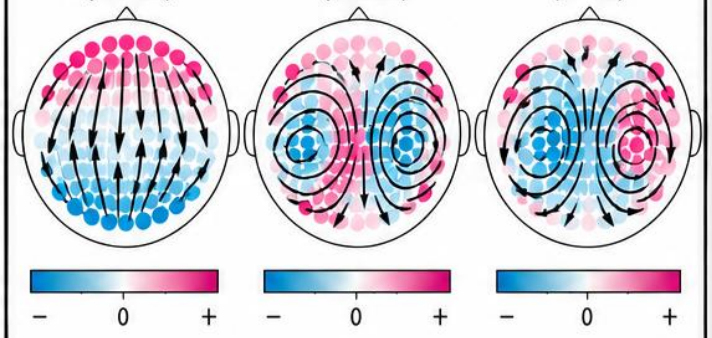
4. Eigenmodes of the 1-Hodge Laplacian

$$\Delta_1 X_k = \lambda_k X_k$$ (X_k : edge flow / tangent vector field)

Gradient mode
(λ small)

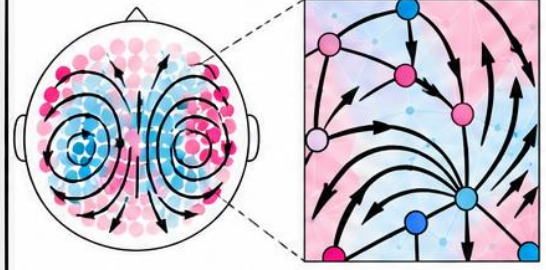
Curl mode
(λ small)

Harmonic mode
($\lambda = 0$)



5. Streamlines on the vector field

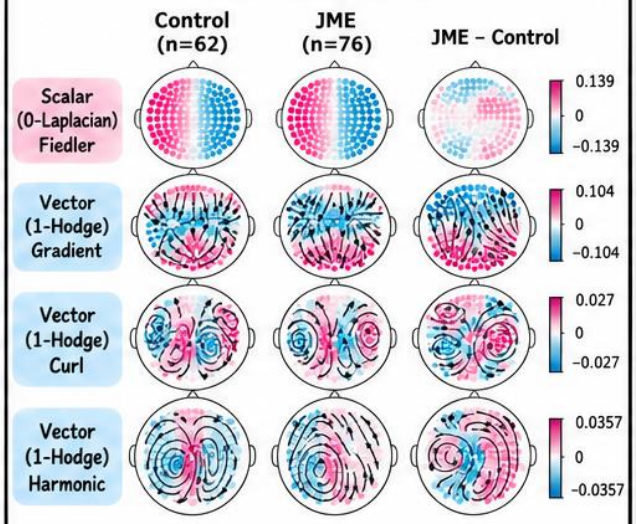
Streamlines (integral curves) show
directions of information flow on the scalp.



Streamlines visualize
recurrent (circular) interactions
in the network.

6. Group comparison of spectral embeddings

Broadband (1–50 Hz)



Speed and acceleration of rapid change in network gradient is related to clinical diagnostic variables

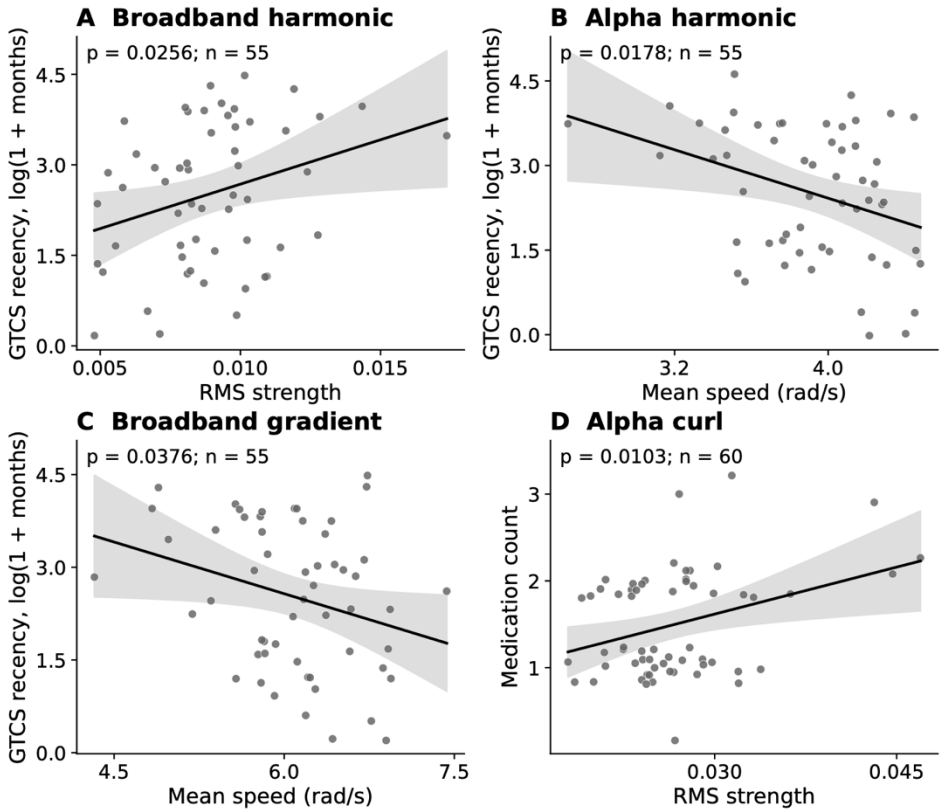


Fig. 3. Selected clinical associations in JME. Points show age/sex-adjusted outcomes; lines and shading show fitted regressions and pointwise 95% confidence intervals. All p-values use one million residual permutations without multiple-comparison adjustment.

***Example* of reporting statistical analysis**

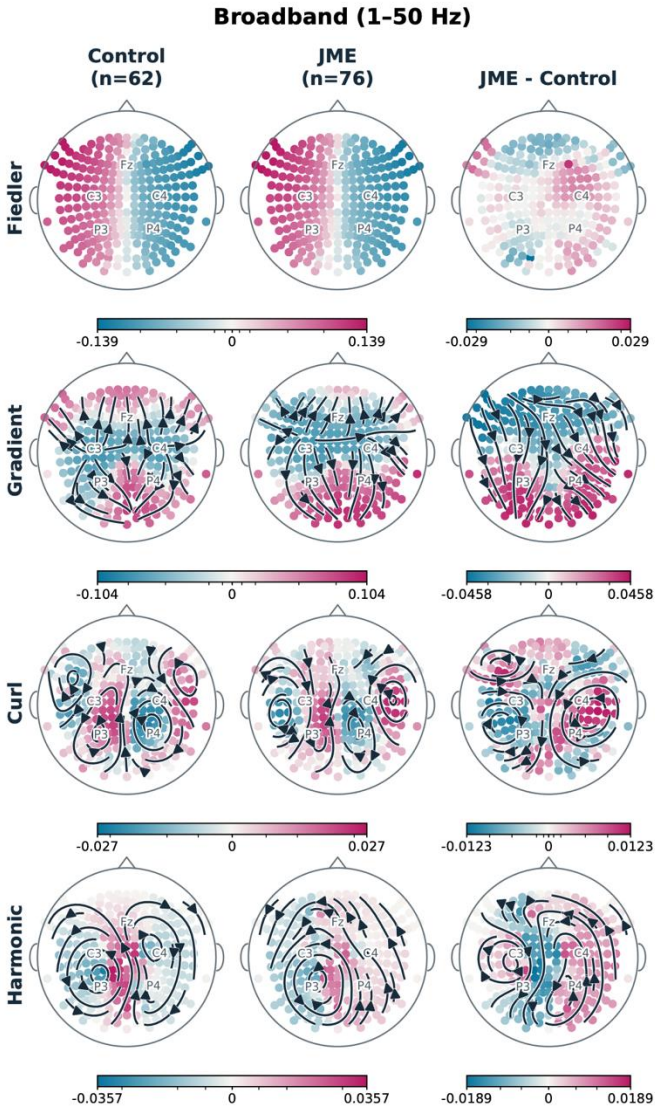


Fig. 1. Broadband (1–50 Hz) group means for controls ($n = 62$) and JME ($n = 76$). Bilateral circulation appears less distinct in the JME harmonic flow, suggesting altered cyclic interactions.

The common component spaces allow flows to be averaged across subjects to estimate population mean patterns. We visualize each group-average edge flow $\bar{X}_i$ as a smoothed planar field, with streamlines defined as integral curves $\mathbf{r}(\tau)$ satisfying [19]:

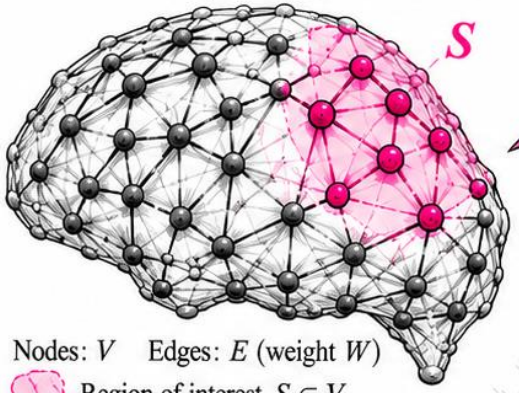
$$\frac{d\mathbf{r}(\tau)}{d\tau} = \bar{X}_i(\mathbf{r}(\tau)), \tag{3}$$

Graph Slepian

Spectrally smooth and spatially localized graph signals

Graph and region of interest

$G = (V, \mathcal{E}, W)$



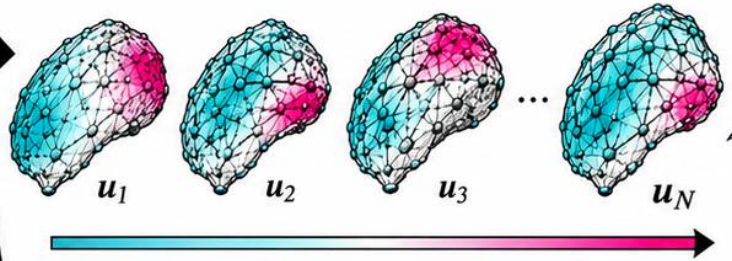
Nodes: V Edges: E (weight W)

Region of interest $S \subset V$

Graph Fourier basis

Graph Laplacian $\boldsymbol{L} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{\mathsf{T}}$

$\boldsymbol{L}\boldsymbol{u}_k = \lambda_k \boldsymbol{u}_k \qquad 0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_N$

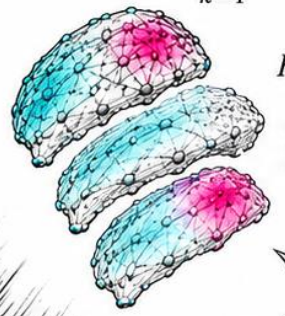


low frequency (smooth) high frequency (oscillatory)

Bandlimited graph signals

Use the first K eigenvectors (smooth signals)

$$\mathbf{g} = \mathbf{U}_K \mathbf{a} = \sum_{k=1}^{K} a_k \boldsymbol{u}_k$$



$K \ll N$

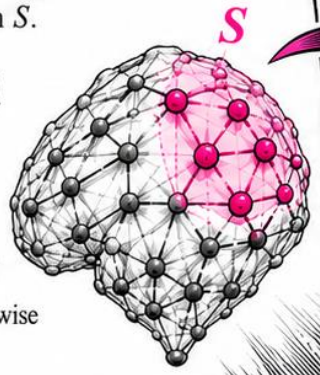
Spatial concentration

Find the bandlimited signal with maximum energy in S.

$$\max_{\mathbf{g}} \frac{\mathbf{g}^{\mathsf{T}} \mathbf{M}_S \mathbf{g}}{\mathbf{g}^{\mathsf{T}} \mathbf{g}}$$

$$\mathbf{g} = \mathbf{U}_K \mathbf{a}$$

$$(M_S)_{ii} = \begin{cases} 1, & i \in S \\ 0, & \text{otherwise} \end{cases}$$

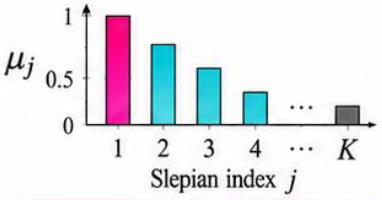


Equivalent eigenproblem

$$\mathbf{C}\mathbf{a} = \mu \mathbf{a}$$

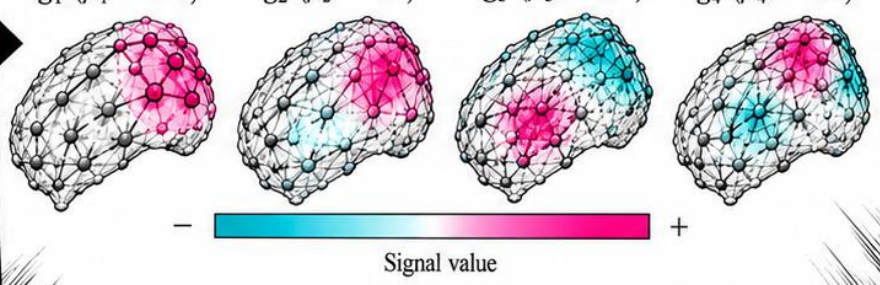
$$\mathbf{C} = \mathbf{U}_K^{\mathsf{T}} \mathbf{M}_S \mathbf{U}_K$$

$0 \le \mu_1 \le \mu_2 \le \cdots \le \mu_K \le 1$

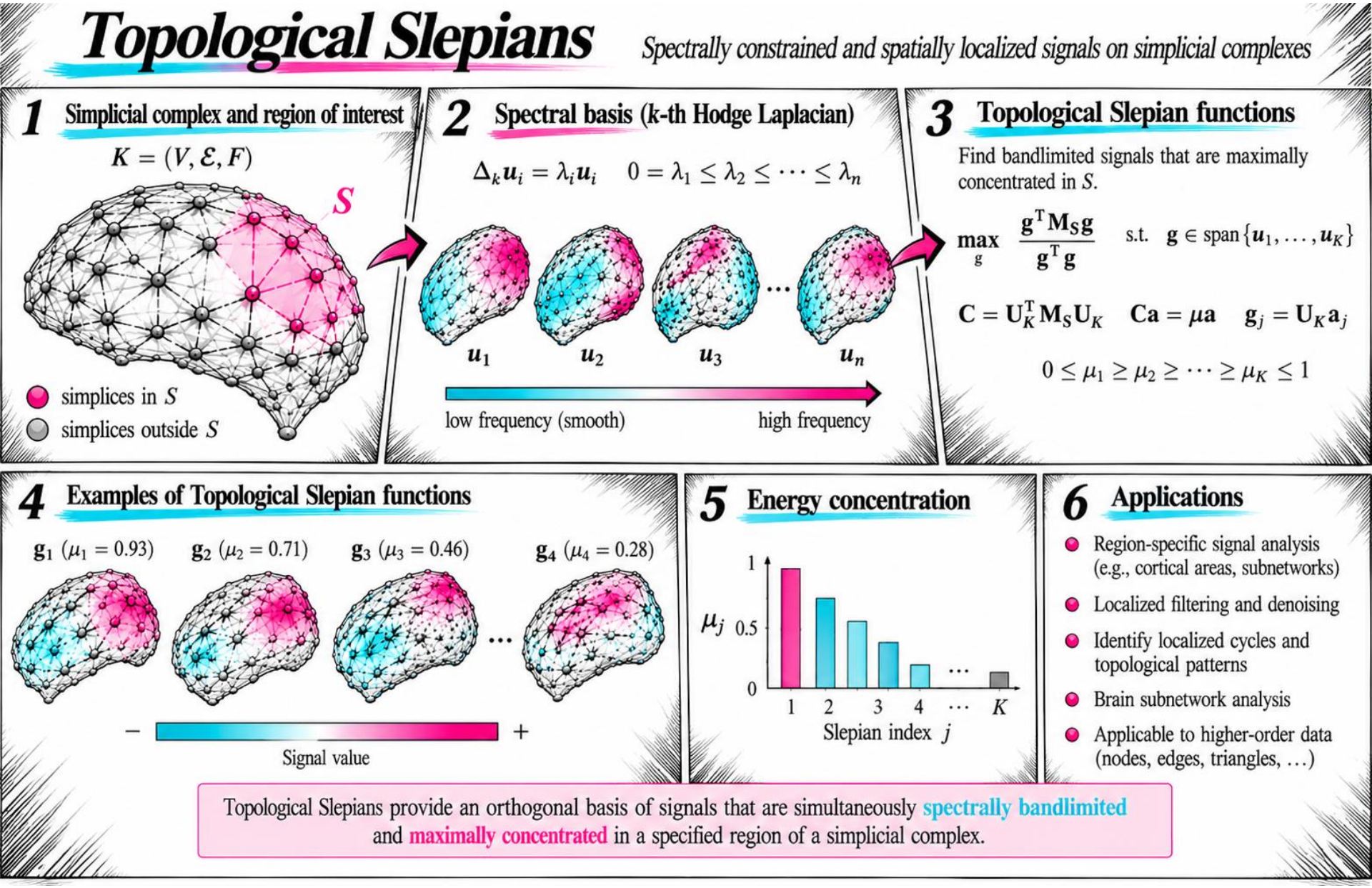


μ_j = fraction of signal energy in S

Graph Slepian functions



Graph Slepian functions provide an orthogonal basis of signals that are simultaneously **smooth** on the graph and **localized** in S.



There are handful of published papers not worth studying.

Project: 1-Hodge Slepian Analysis

Data: EEG epilepsy data, rs-fMRI data

Methods
1. Construction of 1-Hodge Slepian basis along edges
2. Spectral localization and concentration
3. Spectral permutation testing

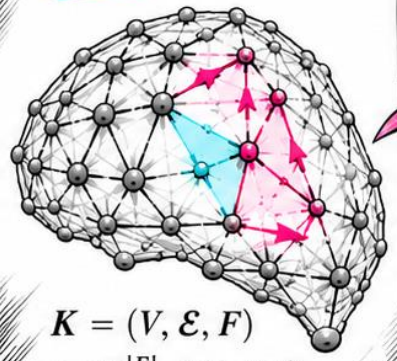
Validation
4. Reproducibility across within and across subjects
5. False-positive and false-negative tests

Application
6. Localization of harmonic cycles in actual data
7. Group-level statistical inference

Hodgelets

Localized multiscale analysis of edge flows on simplicial complexes

1 Simplicial complex and edge flows

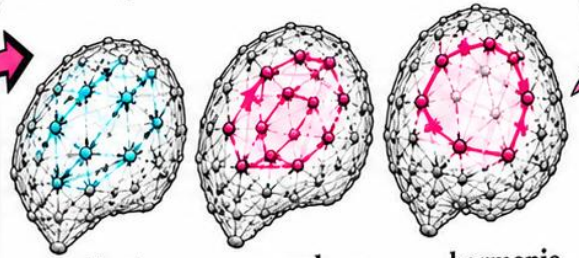


$K = (V, \mathcal{E}, F)$

$x \in \mathbb{R}^{|E|}$ (edge flow)

2 Hodge Laplacian on edges

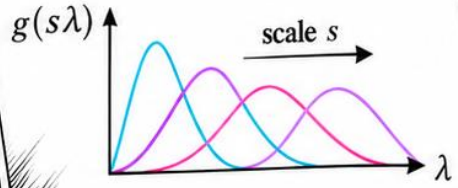
$$\Delta_1 = \mathbf{B}_1^{\mathsf{T}}\mathbf{B}_1 + \mathbf{B}_2\mathbf{B}_2^{\mathsf{T}}$$



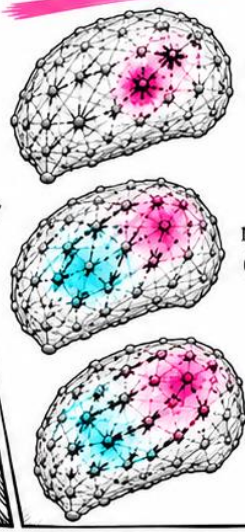
gradient (exact flow) | **curl** (rotational flow) | **harmonic** (cycle flow)

3 Spectral graph wavelets (Hodgelets)

$$\Psi_{s,e} = g(s\Delta_1)\delta_e = \sum_k g(s\lambda_k)\langle \boldsymbol{u}_k, \delta_e\rangle \boldsymbol{u}_k$$



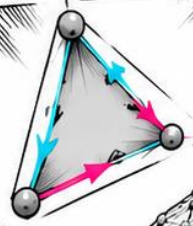
4 Multiscale Hodgelets



small scale (local)

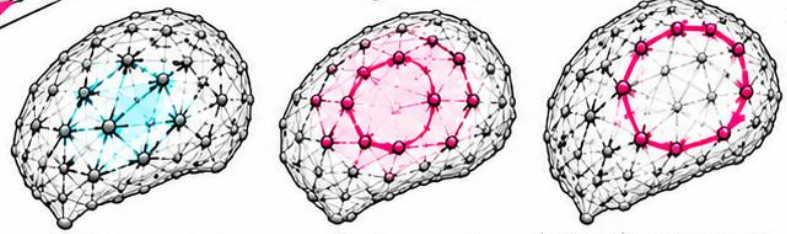
medium scale (regional)

large scale (global)



5 Hodge decomposition in wavelet domain

$$\boldsymbol{x} = \boldsymbol{x}_G + \boldsymbol{x}_C + \boldsymbol{x}_H$$



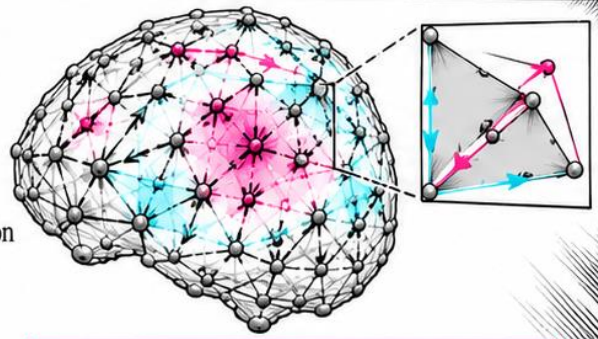
gradient component $\boldsymbol{x}_G \in \mathrm{im}(\mathbf{B}_1^{\mathsf{T}})$

curl component $\boldsymbol{x}_C \in \mathrm{im}(\mathbf{B}_2)$

harmonic component $\boldsymbol{x}_H \in \ker(\Delta_1)$

6 Applications

- Multiscale flow analysis on brain networks
- Localized cycle detection
- Denoising and feature extraction
- Interpret gradient / curl / harmonic organization
- Applicable to higher-order data (nodes, edges, triangles, …)

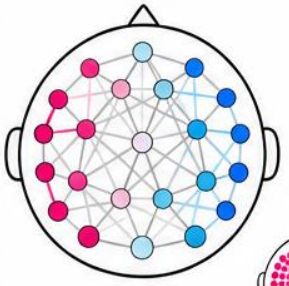


Hodgelets provide a localized, multiscale basis for analyzing edge flows on simplicial complexes.

Sampling in spectral domain

1. Graph and spectral basis

Signals on a graph
(e.g. EEG electrodes)



Graph G
(nodes: electrodes,
edges: functional connections)

Graph Laplacian

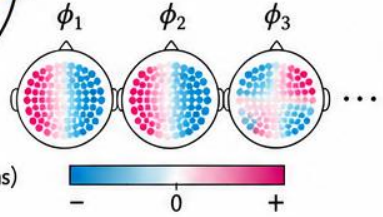
$$L = D - W$$

(D: degree, W: connectivity)

Eigen-decomposition

$$L\phi_k = \lambda_k \phi_k$$

$$0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_N$$



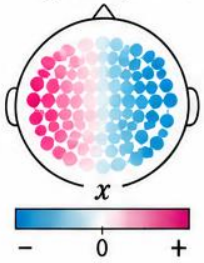
Eigenvectors ϕ_k form a spectral basis
(smooth global patterns on the graph).

2. Spectral representation

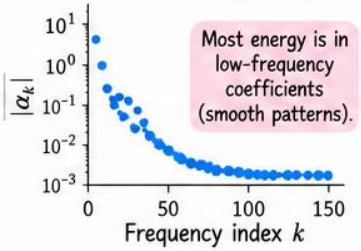
Any signal x on the graph can be expanded in the spectral domain

$$x = \sum_{k=1}^{N} \alpha_k \phi_k \qquad (\alpha_k = \phi_k^T x)$$

Original signal x
(e.g. EEG power)



Spectral coefficients α_k
(in descending order)



Most energy is in low-frequency coefficients (smooth patterns).

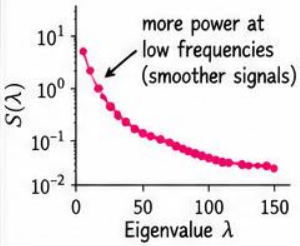
3. Sampling in the spectral domain

Generate a random signal by sampling spectral coefficients.

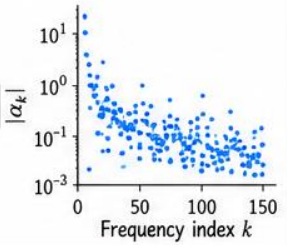
$$\alpha_k \sim \mathcal{N}(0, S(\lambda_k))$$

← spectral power (or empirical spectrum)

Example spectral power $S(\lambda)$



Sampled coefficients α_k



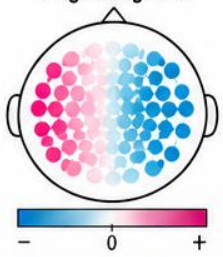
Sampling in the spectral domain:
Draw α_k from a chosen spectrum (e.g. Gaussian with $S(\lambda_k)$).

4. Reconstruct a sample signal

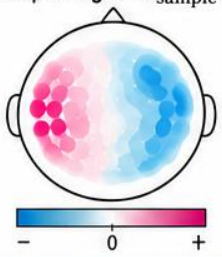
Transform back to the spatial domain using the inverse spectral transform.

$$x_{\text{sample}} = \sum_{k=1}^{N} \alpha_k^{(\text{sample})} \phi_k$$

Original signal x



Sampled signal x_{sample}

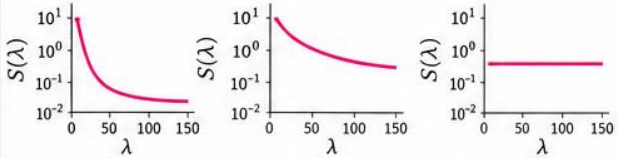


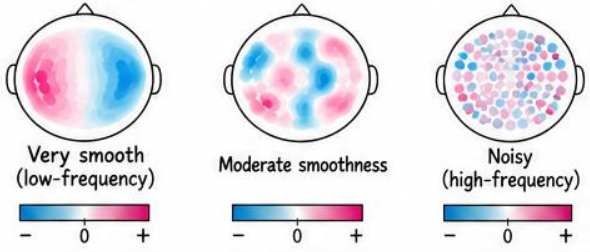
By controlling the spectrum $S(\lambda)$, we can generate realistic signals with desired smoothness.

5. Controlling smoothness by spectrum

Different spectral power $S(\lambda)$ produces signals with different smoothness.

Low-pass (fast decay)	1/f spectrum (moderate decay)	Flat spectrum (white noise)



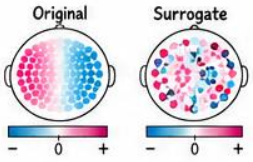


Very smooth (low-frequency) | Moderate smoothness | Noisy (high-frequency)

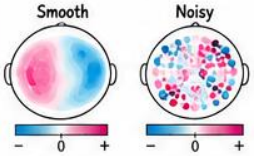
6. Applications

Spectral-domain sampling can be used to:

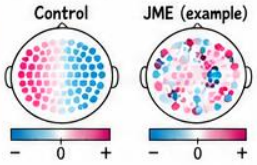
Generate surrogate data
Preserve spectral properties but randomize phases.



Simulate signals with specified smoothness
Control the spectral power $S(\lambda)$.

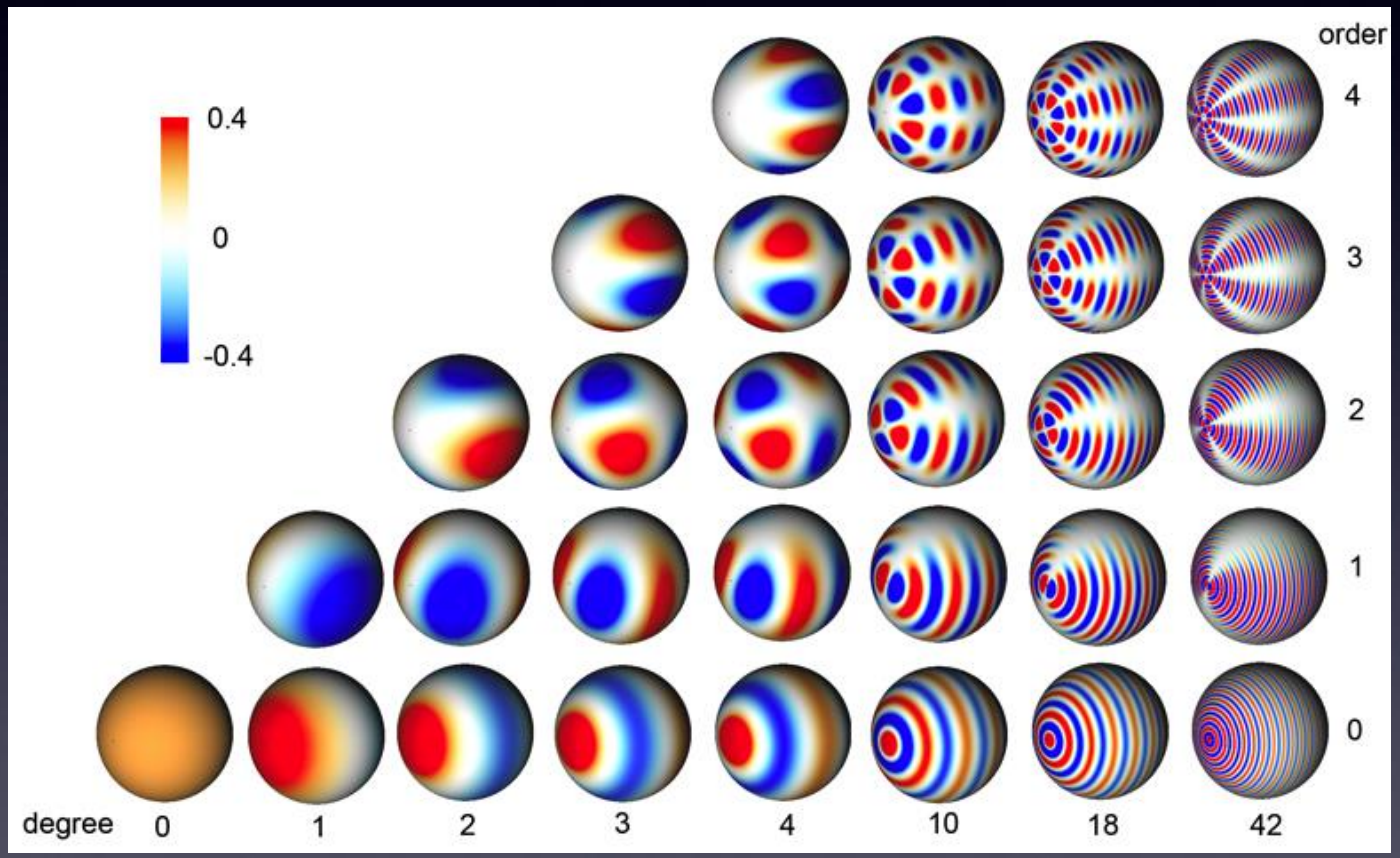


Model group differences
Use different spectra for different groups.



Spherical harmonic of degree *l* and order *m*

$$Y_{lm} = \begin{cases} c_{lm} P_l^{|m|}(\cos\theta)\sin(|m|\varphi), & -l \le m \le -1, \\ \frac{c_{lm}}{\sqrt{2}} P_l^0(\cos\theta), & m = 0, \\ c_{lm} P_l^{|m|}(\cos\theta)\cos(|m|\varphi), & 1 \le m \le l, \end{cases}$$



Coarse detail ⟷ Fine detail



Parameter Estimatio

i-th vertex

$$f(p_i) = \sum_{j=0}^{k} \beta_j \psi_j(p_i)$$

$$\mathbf{f} = (f(p_1), \cdots, f(p_n))'$$

$$\beta = (\beta_0, \cdots, \beta_k)'$$

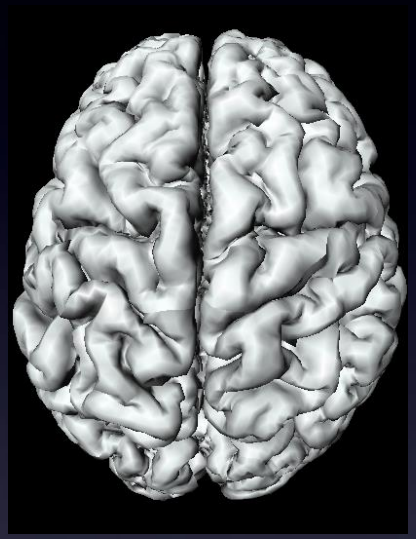
$$\mathbf{Y} = \begin{bmatrix} \psi_0(p_1) & \cdots & \psi_k(p_1) \\ \vdots & \ddots & \vdots \\ \psi_0(p_n) & \cdots & \psi_k(p_n) \end{bmatrix}$$

For *n*>300000
and *k*>(80+1)^2,
it is not so easy to do this
in a laptop with 500MB
memory.

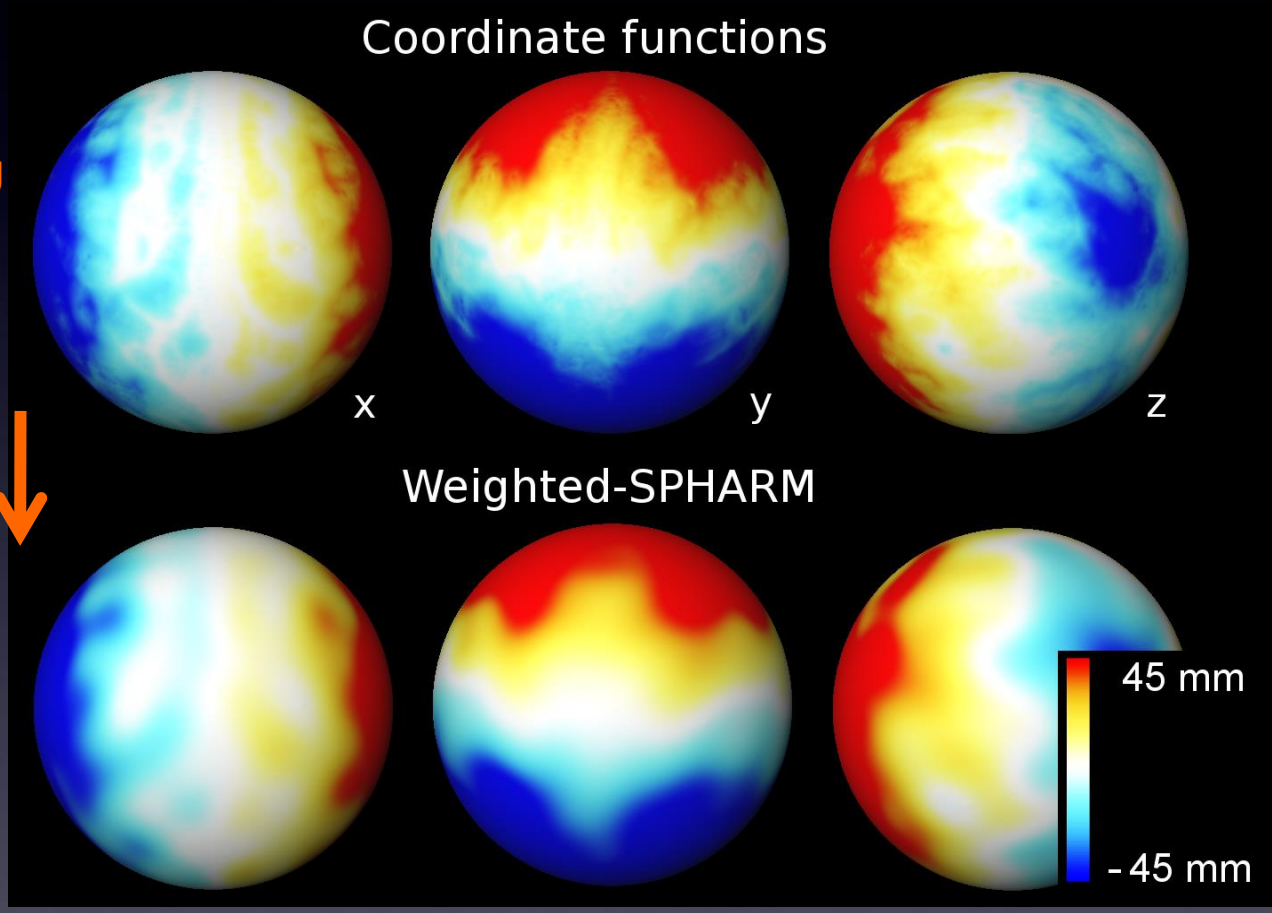
$$\mathbf{f} = \mathbf{Y}\beta$$

$$\beta = (\mathbf{Y}'\mathbf{Y})^{-1}\mathbf{Y}'\mathbf{f}$$

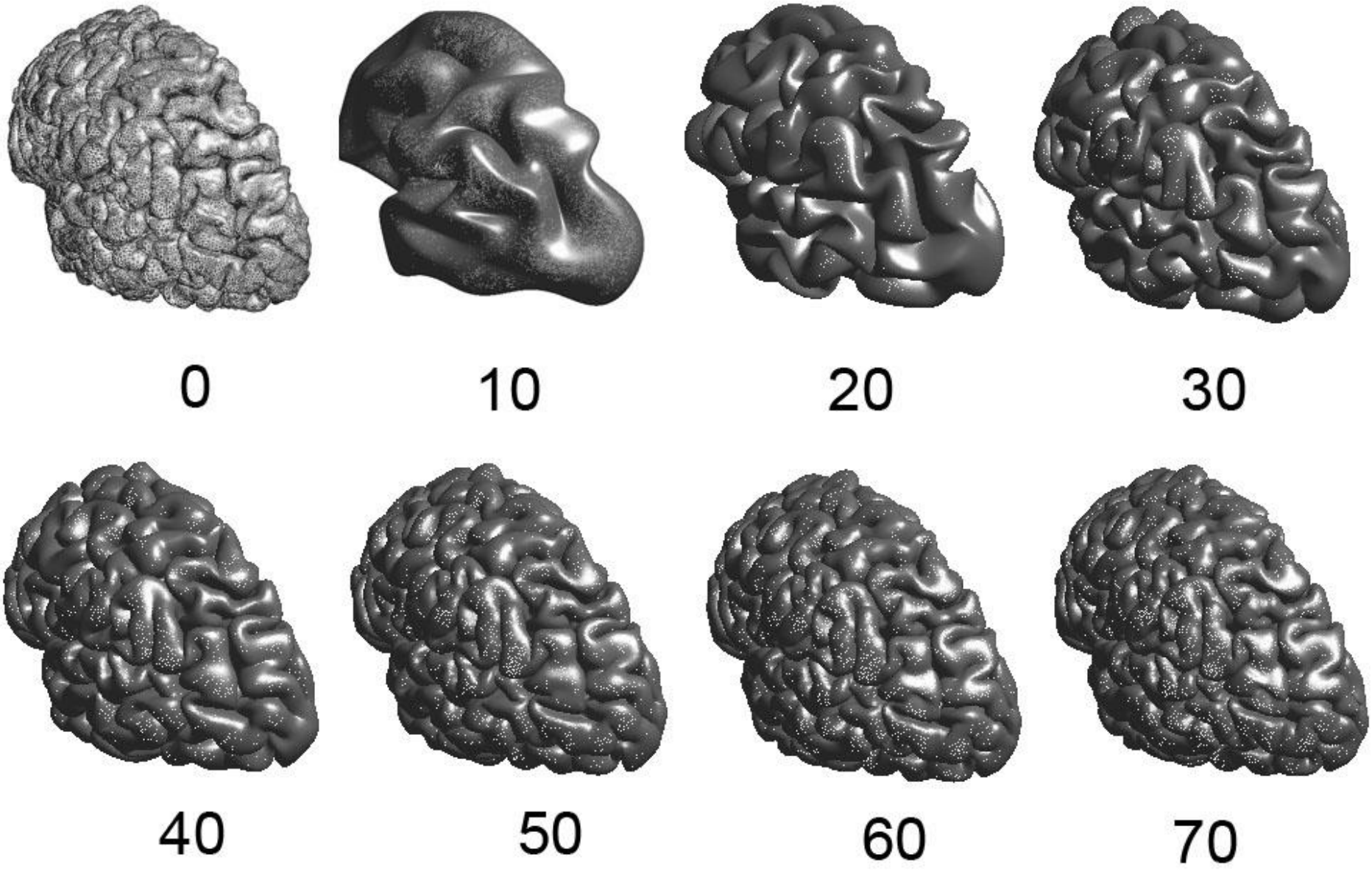
Spherical harmonic representation



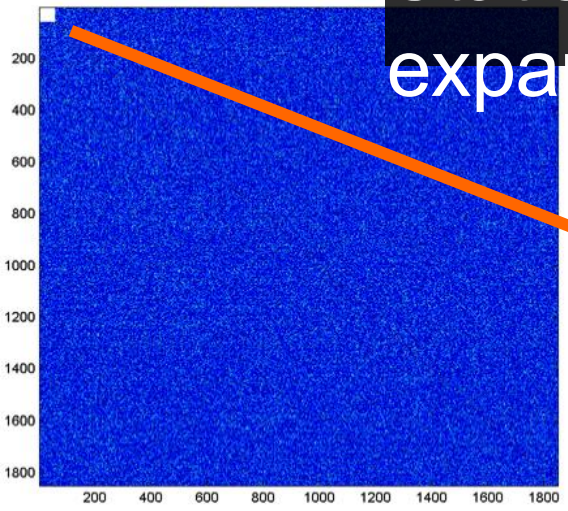
Surface flattening

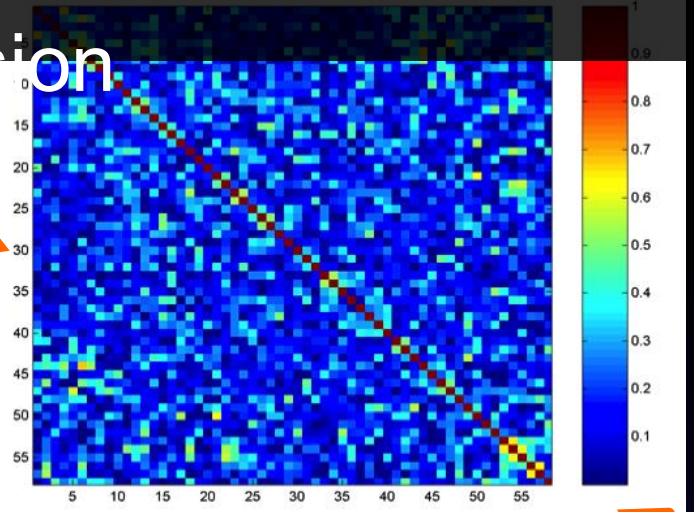


SPAHRM representation of brain surface



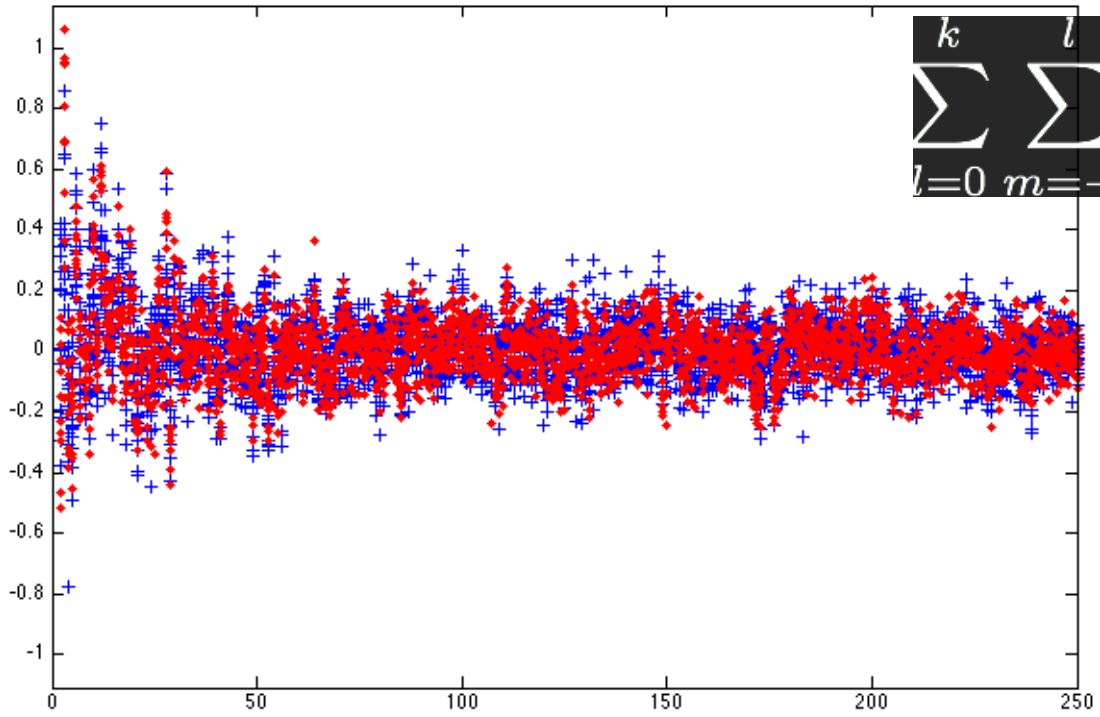
Statistical Model: Karhunen-Loève expansion





uncorrelated normal

Cross correlations



$$\sum_{l=0}^{k} \sum_{m=-l}^{l} e^{-l(l+1)t} \langle f, Y_{lm} \rangle Y_{lm}(\theta, \varphi)$$

Spectral learning

Classification

Discrimination

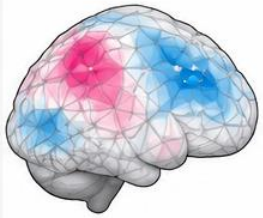
Transfer information

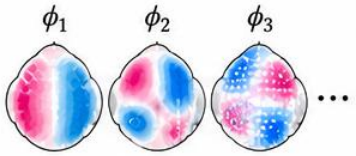
Single-shot spectral transfer learning

1. Spectral representation of signals

Signal on a surface/graph
$x(p)$ (e.g. cortical thickness)

Graph (Laplace-Beltrami) eigenbasis
$-\Delta\phi_k = \lambda_k \phi_k$

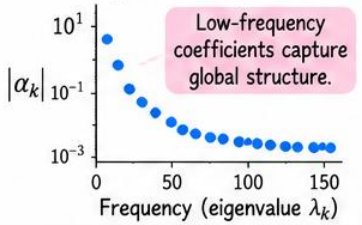




Spectral decomposition

$$x = \sum_{k=1}^{N} \alpha_k \phi_k$$

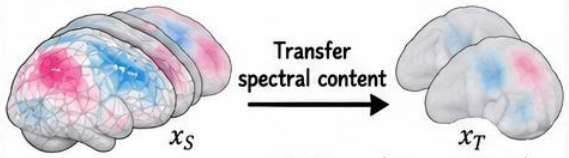
Spectral coefficients

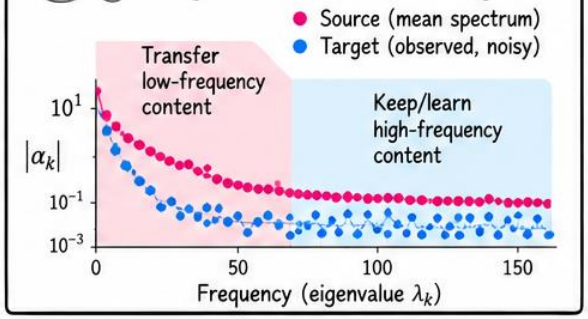


2. Transfer spectral content from source to target

Source domain
(e.g. many subjects)

Target domain
(e.g. few subjects)



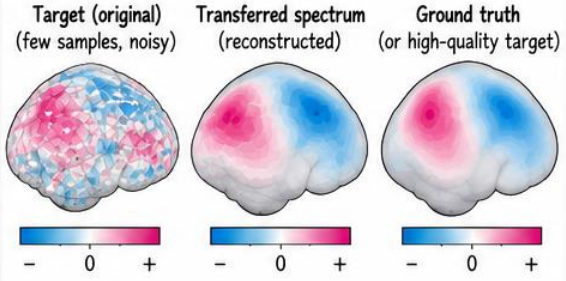


3. Construct target signal with transferred spectrum

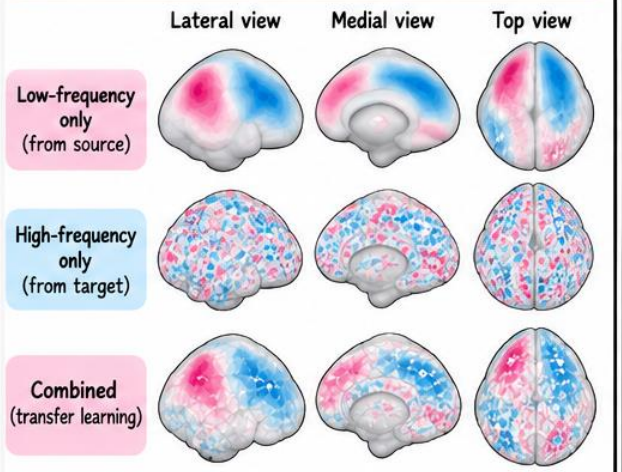
$$\alpha_k^{(T)} = \begin{cases} \alpha_k^{(S)} & (k \le K) \\ \alpha_k^{(T,obs)} & (k > K) \end{cases}$$

Use source spectrum for low frequencies (global structure)

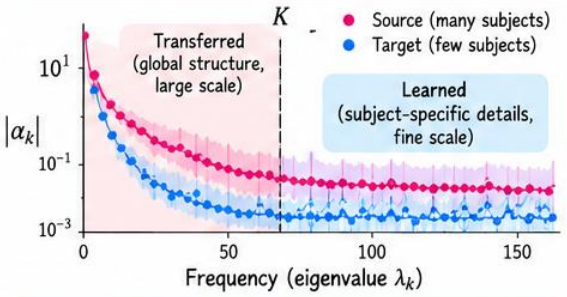
Keep target spectrum for high frequencies (fine details)



4. Effect of spectral transfer



5. Interpretation in the spectral domain

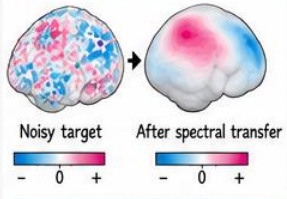


Why it works?

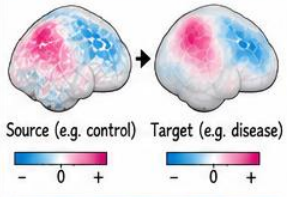
- Low-frequency components capture shared anatomy/structure.
- High-frequency components contain individual or domain-specific details.
- Transferring spectral content provides a strong prior and reduces overfitting when target data are limited.

6. Applications

Denoising / Reconstruction

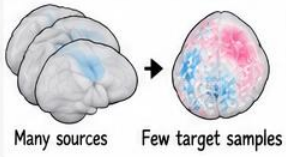


Cross-domain analysis



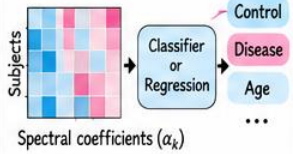
Few-shot learning

Learn from many subjects, adapt to a new subject



Classification / Prediction

Use transferred spectral features



Project: Single shot transfer learning through Hodge Laplacian spectrum